

HYPERLIQUID <> LIGHTER RWA PRICE GAP ANALYSIS

Report Generated: 2026-04-25 16:22:18

Repository: C:\Users\user\Desktop\MoneyLaundering\Hyper-LighterARB

Data Range: 2026-04-23 08:17:56+00:00 to 2026-04-25 07:49:56+00:00

Assets: BRENT OIL, GOLD, SILVER, WTI

EXECUTIVE SUMMARY

Raw Matched Seconds: 93,433

Fresh Matched Seconds Used: 92,079

Rows Removed by Freshness Filter: 1,354 (1.4%)

Freshness Rule: $\text{abs}(\text{captured_at_utc} - \text{exchange_time}) \leq 2\text{s}$ on both venues

PRICE GAP SUMMARY

BRENT OIL	mean= 1.58 bps median= 1.80 bps p90_abs= 3.44 bps richer=Hyperliquid crosses=1520
GOLD	mean= 1.91 bps median= 2.06 bps p90_abs= 3.63 bps richer=Hyperliquid crosses= 238
SILVER	mean= -2.37 bps median= -1.96 bps p90_abs= 5.73 bps richer=Lighter crosses=1087
WTI	mean= -2.36 bps median= -2.48 bps p90_abs= 5.17 bps richer=Lighter crosses=1160

STRICT PAIR-TRADE TEST

Entry: $|\text{mid gap}| \geq 8.0 \text{ bps}$, executable edge $\geq 5.0 \text{ bps}$

Depth: depth-5 capacity $\geq 100, \text{maxnotional}/10,000$

Exit: gap $\leq 0.25 \text{ bps}$, price cross, or 600s max hold

Fees: 1.0 bps per leg per fill, four fills per pair trade

Realized Trades: 15

Gross PnL: \$55.00

Fees: \$20.80

Net PnL: \$34.21

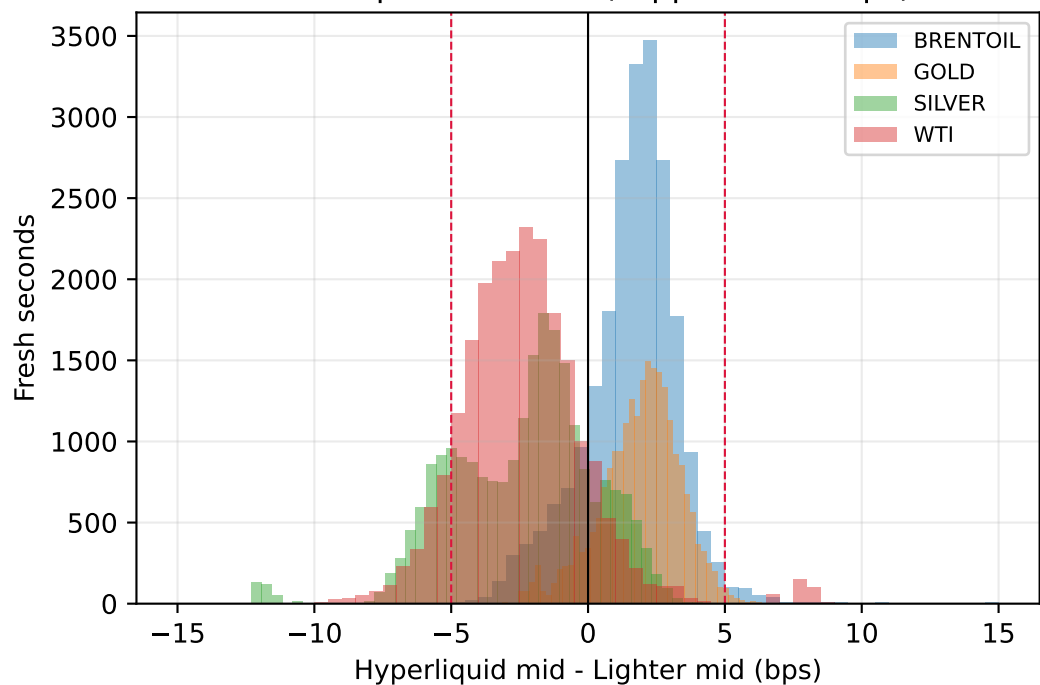
Net Win Rate: 86.7%

Avg Holding Period: 82.8s

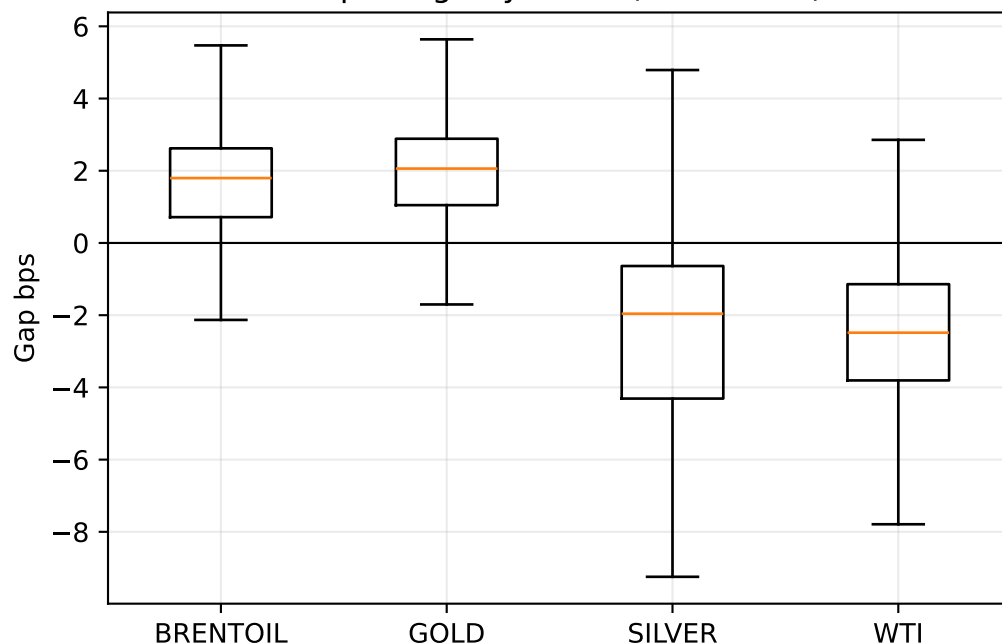
Avg Holding Period for Profitable Trades: 47.7s

Price Gap Distribution Analysis

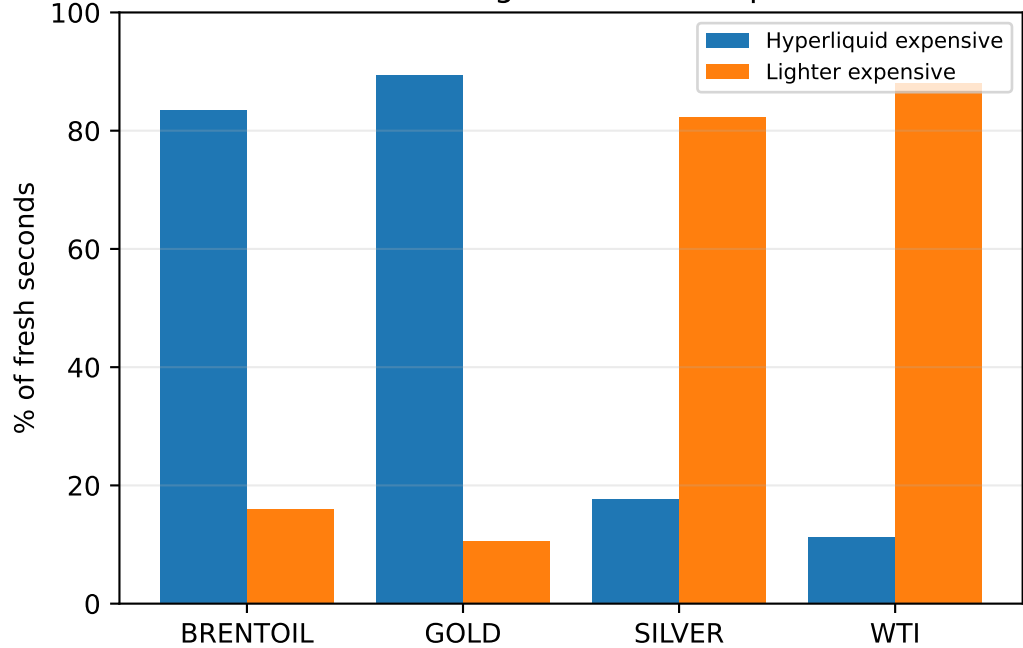
Mid Gap Distribution (clipped +/-15 bps)



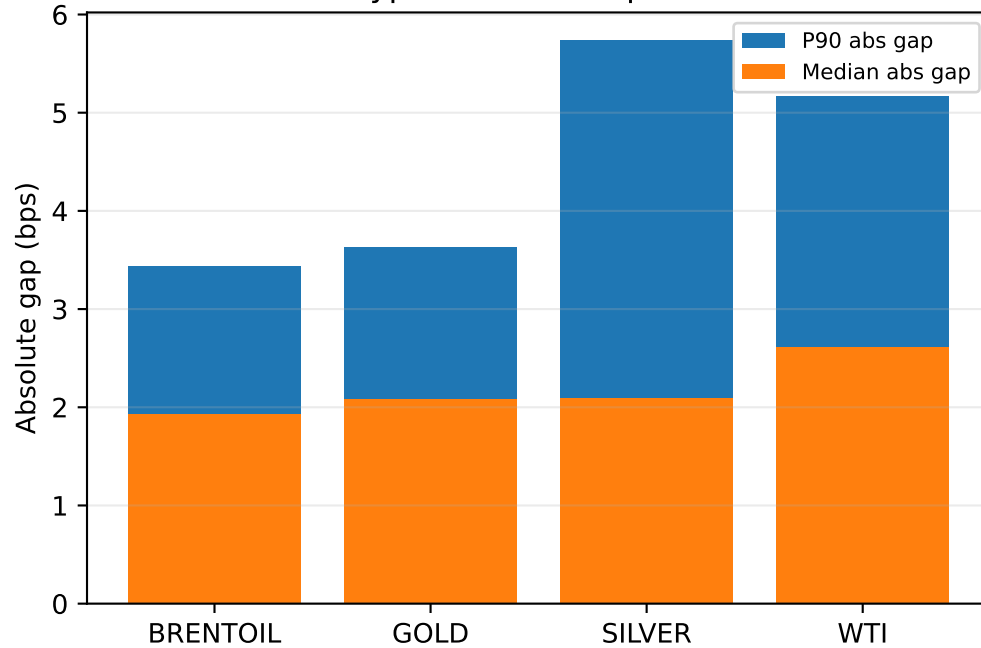
Gap Range by Asset (no outliers)



Which Exchange Was More Expensive

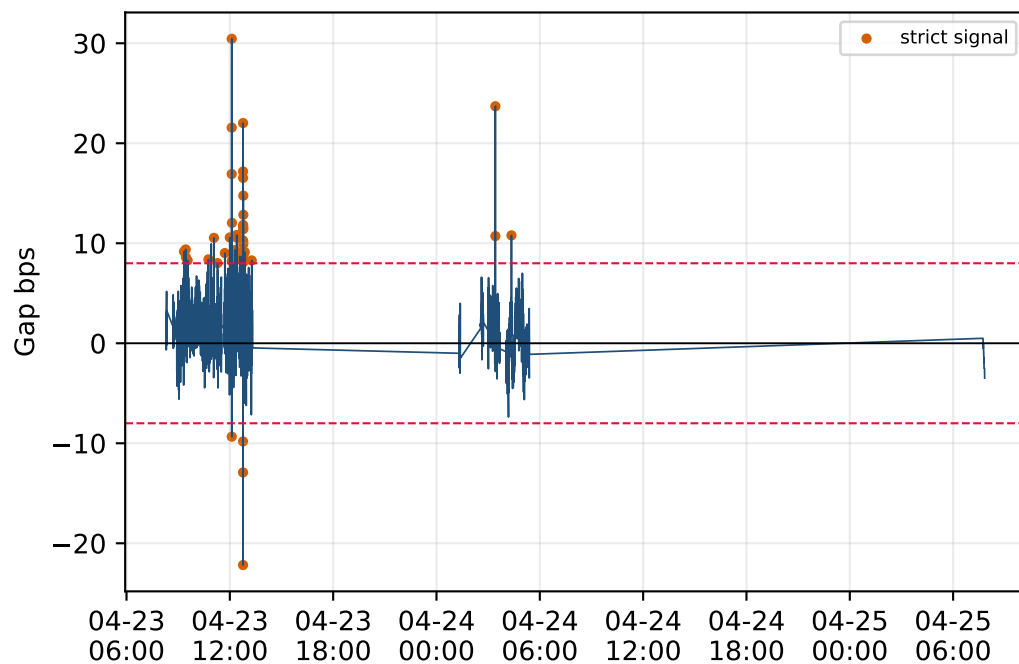


Typical vs Tail Gap Size

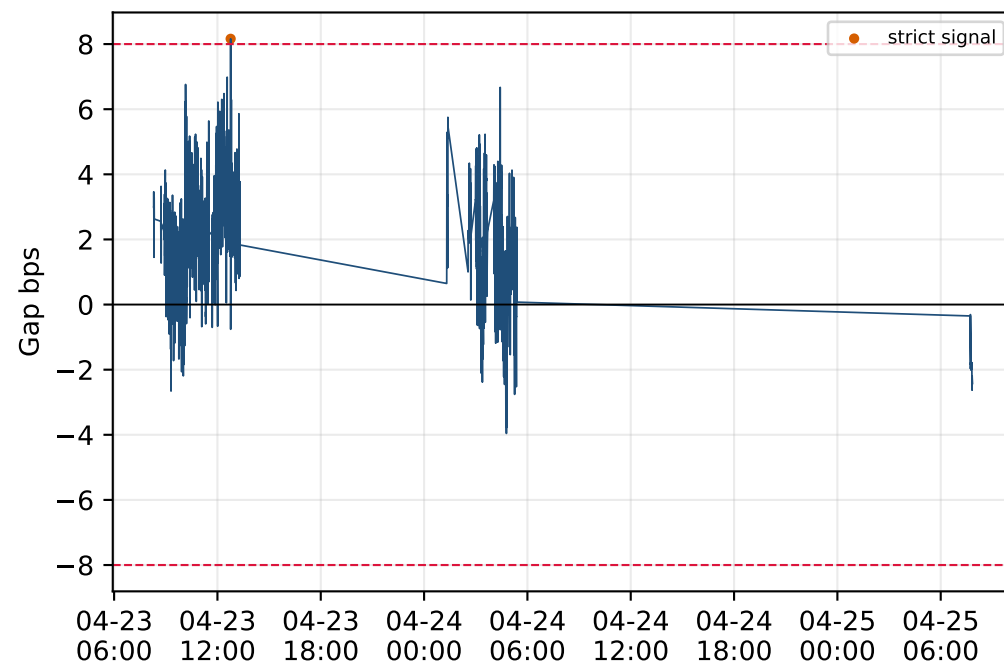


Timing and Clustering Analysis

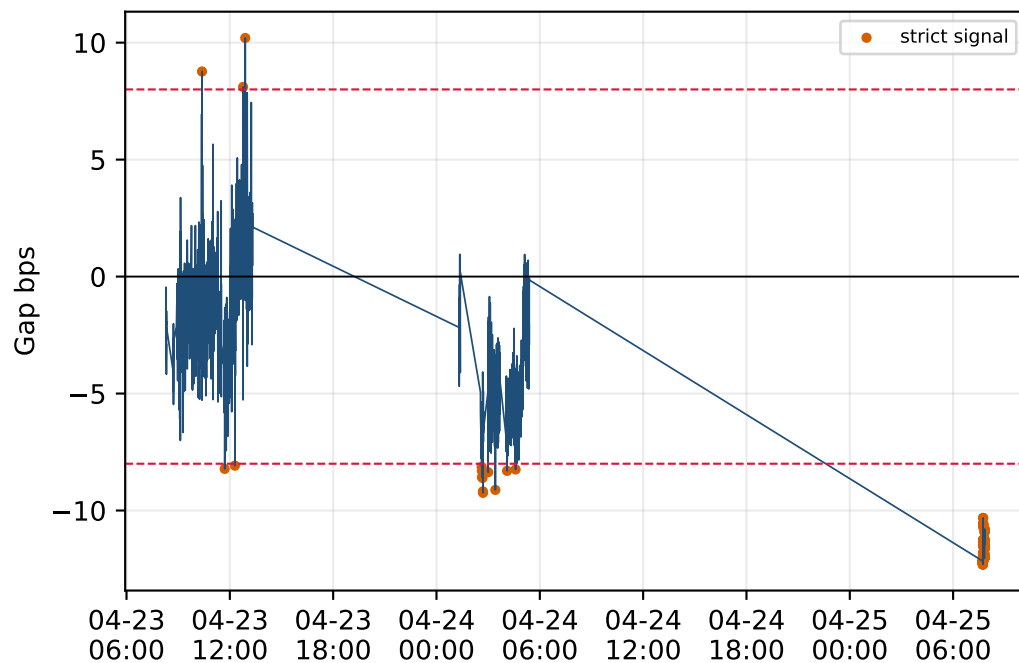
BRENTOIL



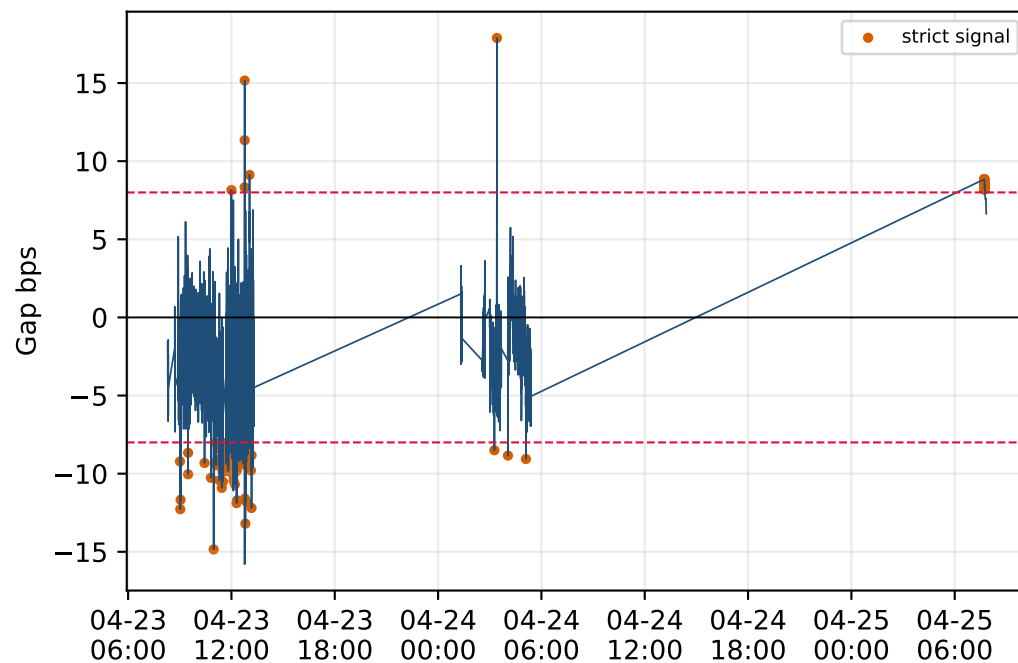
GOLD



SILVER

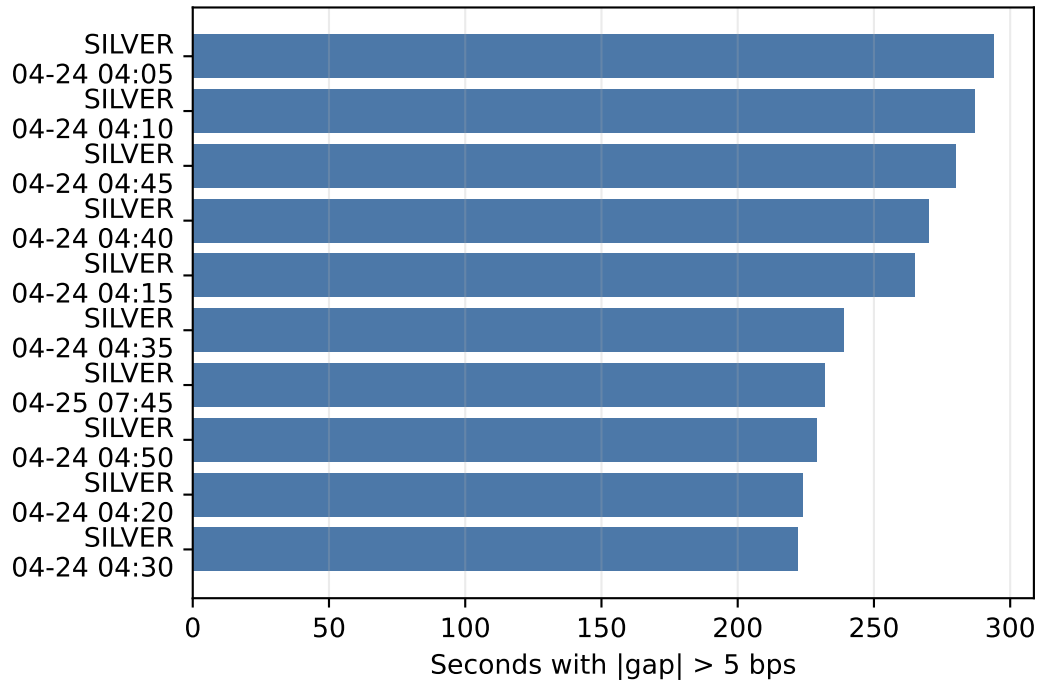


WTI

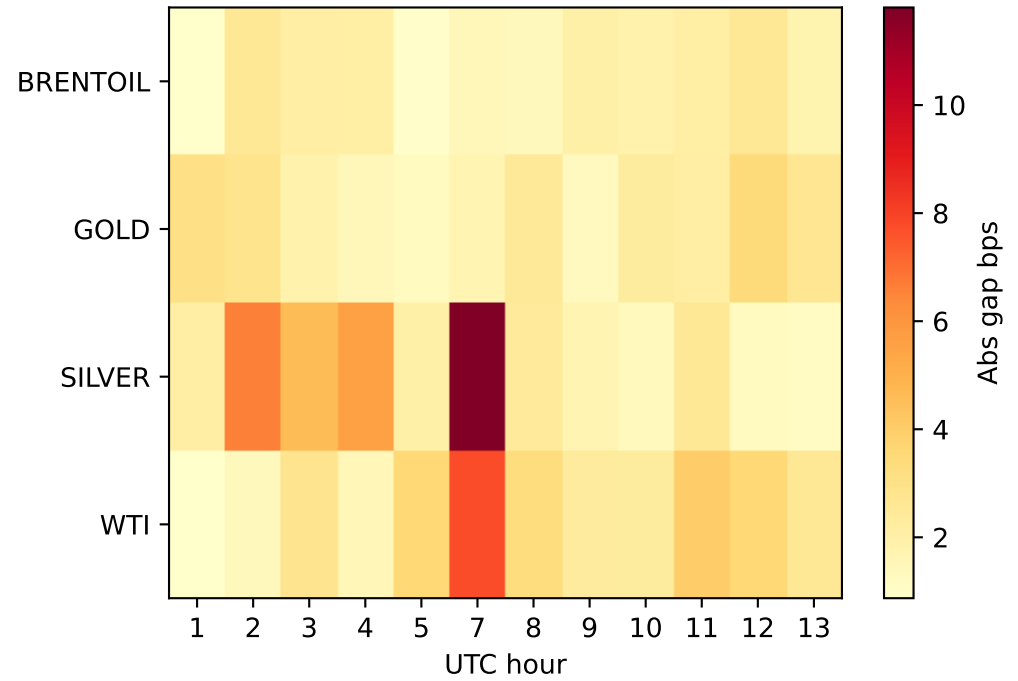


Convergence and Pair-Trade Analysis

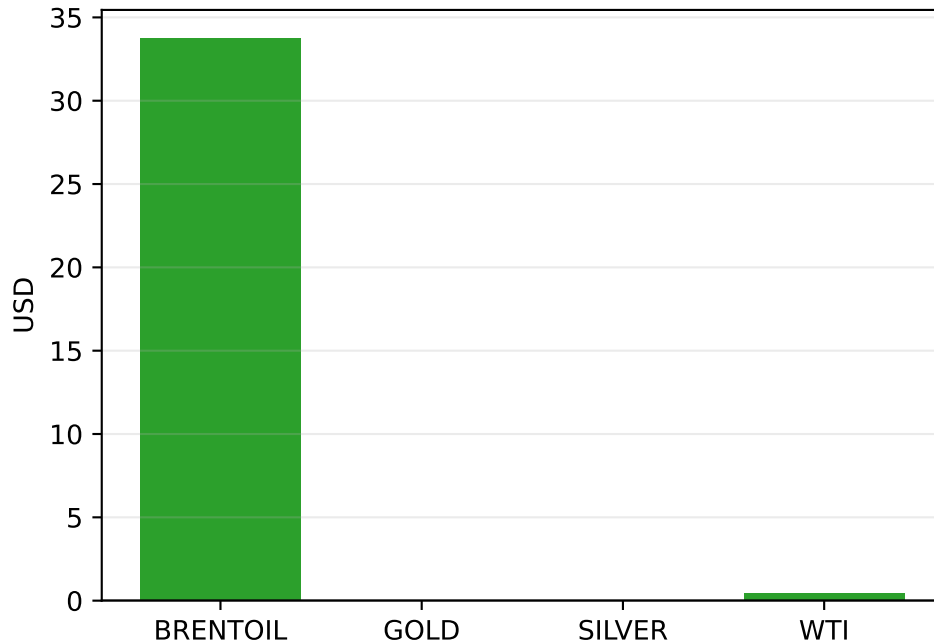
Top 5-Minute Gap Clusters



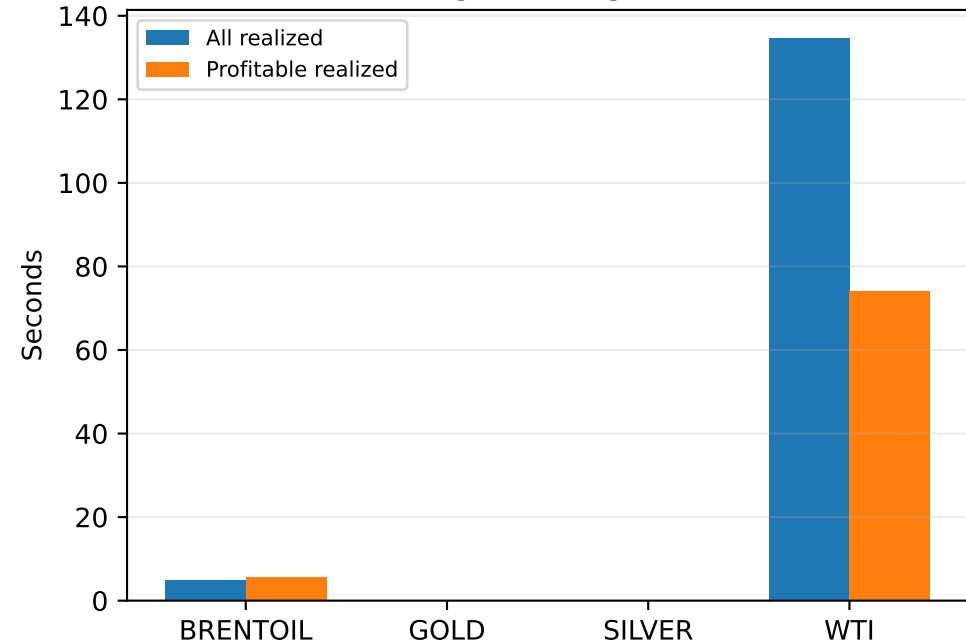
Mean Absolute Gap by UTC Hour



Strict Taker Backtest Net PnL



Average Holding Period



PAIR-TRADE STRATEGY ANALYSIS

EXCHANGE RICHNESS

BRENT0IL Hyperliquid was richer 83.5% of fresh seconds.

GOLD Hyperliquid was richer 89.4% of fresh seconds.

SILVER Lighter was richer 82.2% of fresh seconds.

WTI Lighter was richer 88.1% of fresh seconds.

CONVERGENCE / CROSSING BEHAVIOR

BRENT0IL zero-cross count=1520, |gap|>5bps= 1.7%, |gap|>8bps= 0.3%.

GOLD zero-cross count= 238, |gap|>5bps= 0.8%, |gap|>8bps= 0.0%.

SILVER zero-cross count=1087, |gap|>5bps= 17.9%, |gap|>8bps= 1.5%.

WTI zero-cross count=1160, |gap|>5bps= 11.4%, |gap|>8bps= 1.1%.

TIME CLUSTERING

SILVER 2026-04-24 04:05 UTC: 294s above 5bps, max_abs= 8.31bps, strict_signal_seconds=1.

SILVER 2026-04-24 04:10 UTC: 287s above 5bps, max_abs= 7.17bps, strict_signal_seconds=0.

SILVER 2026-04-24 04:45 UTC: 280s above 5bps, max_abs= 7.84bps, strict_signal_seconds=0.

SILVER 2026-04-24 04:40 UTC: 270s above 5bps, max_abs= 7.98bps, strict_signal_seconds=0.

SILVER 2026-04-24 04:15 UTC: 265s above 5bps, max_abs= 7.01bps, strict_signal_seconds=0.

SILVER 2026-04-24 04:35 UTC: 239s above 5bps, max_abs= 8.25bps, strict_signal_seconds=1.

SILVER 2026-04-25 07:45 UTC: 232s above 5bps, max_abs=12.10bps, strict_signal_seconds=232.

SILVER 2026-04-24 04:50 UTC: 229s above 5bps, max_abs= 6.85bps, strict_signal_seconds=0.

STRICT BACKTEST BY ASSET

BRENT0IL trades= 6, net=\$ 33.76, win= 83.3%, avg_hold= 5.0s, profitable_avg_hold= 5.6s.

WTI trades= 9, net=\$ 0.45, win= 88.9%, avg_hold= 134.7s, profitable_avg_hold= 74.0s.

KEY OBSERVATIONS

1. Most gaps are small: the normal regime is roughly 2-6 bps, so taker fees matter.
2. Hyperliquid was structurally richer in BRENT0IL and GOLD; Lighter was richer in SILVER and WTI.
3. The strongest strict realized opportunities came from BRENT0IL around 2026-04-23 12:45 UTC.
4. WTI showed frequent convergence but poor displayed Hyperliquid capacity, so size must be capped.
5. SILVER had large late-sample Lighter-rich clusters, but the sample ended before proving realized

UNRESOLVED SIGNAL WARNING

SILVER 2026-04-25 long_hyper_short_lighter: 321s of signals, 07:43:07-07:49:54 UTC.

WTI 2026-04-23 long_hyper_short_lighter: 116s of signals, 09:00:36-13:10:08 UTC.

WTI 2026-04-25 short_hyper_long_lighter: 110s of signals, 07:43:07-07:45:22 UTC.